

SAJC JC2 Preliminary Exam Paper 1 Worked Solutions

1. Volume of CO_2 removed by aqueous $\text{NaOH} = 75 - 25 = 50 \text{ cm}^3$
Volume of excess $\text{O}_2 = 25 \text{ cm}^3$
Volume of O_2 reacted $= 100 - 25 = 75 \text{ cm}^3$
- $$\text{C}_x\text{H}_y + \text{O}_2 \rightarrow x\text{CO}_2 + \frac{y}{2}\text{H}_2\text{O}$$
- From the ratio of gases, since 10 cm^3 of hydrocarbon produced 50 cm^3 of CO_2 , $x = 5$
Since 10 cm^3 of hydrocarbon reacted with 75 cm^3 of O_2 :
- $$\text{C}_5\text{H}_y + \frac{15}{2}\text{O}_2 \rightarrow 5\text{CO}_2 + \frac{y}{2}\text{H}_2\text{O}$$
- $y = 10$
Therefore, molecular formula of hydrocarbon $= \text{C}_5\text{H}_{10}$
- Ans: C** C_5H_{10}

2. $\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$
 $\text{Fe}^{2+} \rightarrow \text{Fe}^{3+} + \text{e}^-$
Mole ratio of $\text{Fe}^{2+} : \text{MnO}_4^- = 5 : 1$
- $$\text{MnO}_4^- + 4\text{H}^+ + 3\text{e}^- \rightarrow \text{MnO}_2 + 2\text{H}_2\text{O}$$
- Mole ratio of $\text{Fe}^{2+} : \text{MnO}_4^- = 3 : 1 = 5 : \frac{5}{3}$
- Therefore, volume of MnO_4^- solution needed to oxidise same amount of Fe^{2+} in neutral condition $= \frac{5}{3} \times 15.0 = 25.0 \text{ cm}^3$
- Ans: C** 25.0 cm^3

3. Option **A** – Maximum oxidation number of element in oxide across period 2 increases from +1 to +4 from Li_2O to CO_2 . Maximum oxidation number of oxygen in oxide (O_2) is zero.

Option **B** – Refer to data booklet. First ionisation energy dips at boron, and again at oxygen.

Option **C** – Across the period, nuclear charge increases as proton number increases.

Option **D** – Across the period, nuclear charge increases, but increase in shielding effect is insignificant, hence effective nuclear charge increases and atomic radius decreases across period 2.

Ans: **C** Nuclear charge

4. Ideal gas equation: $pV = nRT$

Option **A** – Correctly describes behaviour.

At constant p , $\frac{V}{T}$ is constant. Graph reflects this as a horizontal straight line.

Option **B** – Correctly describes behaviour.

At constant T , pV is constant. $\rightarrow p \propto \frac{1}{V}$.

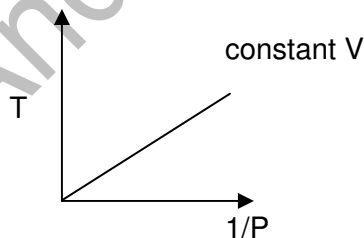
Option **C** – Incorrectly describes behaviour.

At constant V , $T \propto p$. If graph is a straight line passing through origin, the x-axis should be labelled as “P” instead.

Option **D** – Correctly describes behaviour.

At constant T , pV is constant. Regardless of V , pV is a fixed value.

Ans: **C**



5. Option **A** – False.

To act as a ligand, phosphorus atom must have a lone pair of electrons. All the phosphorus atoms in P_4O_{10} do not have lone pair of electrons.

Option **B** – False.

One of the phosphorus atoms in P_4O_9 has a lone pair of electrons.

Option **C** – False.

If the phosphorus atoms are not identical, i.e. bonded to different number of oxygen atoms, they do not have the same oxidation state.

Option **D** – True.

All the phosphorus atoms have 4 electron pairs around it, either as bonding pairs or as lone pairs.

Ans: **D** The phosphorus atoms of both oxides have electron pairs that are arranged tetrahedrally.

6. Option **A**: Giant covalent, ionic, giant covalent
 Option **B**: Simple covalent, simple covalent, giant covalent
 Option **C**: Simple covalent, ionic, giant covalent
 Option **D**: Simple covalent, simple covalent, giant covalent

Ans: **C** Water, lithium aluminium hydride, diamond

7. Option **A**: Bond formation is exothermic process.

Option **B**: Temperature of the water drops, indicating enthalpy change of solution is endothermic.

Option **C**: $\Delta H_{\text{sol}} = \Delta H_{\text{hyd}} - LE$
 $= -ve - (-ve)$

Hence LE must be more exothermic than ΔH_{hyd} to make ΔH_{sol} +ve

Option **D**: $NH_4NO_3(s) \rightarrow NH_4NO_3(aq)$ ΔH_{sol}

$N_2(g) + 2H_2(g) + 3/2 O_2(g) \rightarrow NH_4NO_3(s)$ ΔH_f

The 2 equations are different. Hence not enough information to conclude enthalpy change of formation.

Ans: **C** The lattice energy of ammonium nitrate is more exothermic than the

hydration energy of the ions.

$$\begin{aligned}
 8. \quad \Delta H_r &= 3\Delta H_f(\text{CO}) + 2\Delta H_f(\text{Fe}) - \Delta H_f(\text{Fe}_2\text{O}_3) - 3\Delta H_f(\text{C}) \\
 &= 3(-110) + 2(0) - (-825) - 3(0) \\
 &= +495 \text{ kJ mol}^{-1}
 \end{aligned}$$

Ans: C +495 kJ mol⁻¹

9. Comparing experiments 1 & 2, keeping P_{NO} constant,
When P_{H2} is doubled, rate is doubled. Order w.r.t. H₂ is 1.

Comparing experiments 2 & 3, keeping P_{H2} constant,
When P_{NO} is doubled, rate is quadrupled. Order w.r.t NO is 2.

$$\text{Rate} = k[\text{H}_2][\text{NO}]^2 \quad \text{Overall order of reaction} = 3$$

Comparing experiments 1 & 4,

$$\frac{\text{Rate}_1}{\text{Rate}_4} = \frac{1}{x} = \frac{k(0.20)^2(0.40)}{k(0.80)^2(0.20)}$$

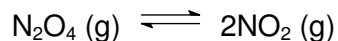
$$x = 8$$

Ans: C The value of x is 8.

$$\begin{aligned}
 10. \quad \left(\frac{180}{1800}\right) &= \left(\frac{1}{2}\right)^n \\
 n \times t_{1/2} &= 1 \times 7 \times 24 \\
 t_{1/2} &= 50 \text{ h}
 \end{aligned}$$

Ans: C 50 h

11. Partial pressure of NO₂ = mole fraction of NO₂ × total pressure, p
Assume initial no. of mole of N₂O₄ to be 1. Since degree of dissociation is y ,



	N ₂ O ₄ (g)	2NO ₂ (g)
Initial / no. of mol	1	0
Change / no. of mol	-y	+2y
Equilibrium /no. of mol	1-y	2y

$$\text{Total no. of mol} = 1 - y + 2y = 1 + y$$

$$\text{Partial pressure of NO}_2 = [2y / (1+y)] \times p$$

Ans: D $\left(\frac{2y}{1+y}\right)p$

- 12.** This is a strong acid – weak base titration, when 12.5cm³ of weak base added. We have excess of strong acid, just calculate no. of mol of excess acid and from there calculate [H⁺] and then pH.

$$\begin{aligned} \text{When 12.5cm}^3 \text{ of CH}_3\text{NH}_2 \text{ added, no. of mol of CH}_3\text{NH}_2 &= (12.5 / 1000) \times 0.1 \\ &= 1.25 \times 10^{-3} \text{ mol} \end{aligned}$$

$$\text{No. of mol of HCl reacted} = 1.25 \times 10^{-3} \text{ mol}$$

$$\text{No. of mol of HCl remaining} = (25/1000) \times 0.1 - (1.25 \times 10^{-3}) = 1.25 \times 10^{-3} \text{ mol}$$

$$[\text{HCl}] = [\text{H}^+] = (1000/37.5) \times (1.25 \times 10^{-3}) = 0.0333 \text{ mol dm}^{-3}$$

$$\text{pH} = -\lg [\text{H}^+] = 1.48$$

Ans: A 1.48

- 13.** No. of mol of OH⁻ added = 2/1000 × 0.5 = 1 × 10⁻³ mol
No. of mol of Mg²⁺ = No. of mol of Ca²⁺ = 0.2 mol

$$\begin{aligned} \text{I.P. of Mg(OH)}_2 &= [\text{Mg}^{2+}][\text{OH}^-]^2 = (1000/1002 \times 0.2)(1000/1002 \times 1 \times 10^{-3})^2 \\ &= 1.98 \times 10^{-7} \text{ mol}^3 \text{ dm}^{-9} > K_{\text{sp}} \text{ of Mg(OH)}_2 \end{aligned}$$

=> Mg(OH)₂ ppt formed

$$\begin{aligned} \text{I.P. of Ca(OH)}_2 &= [\text{Ca}^{2+}][\text{OH}^-]^2 = (1000/1002 \times 0.2)(1000/1002 \times 1 \times 10^{-3})^2 \\ &= 1.98 \times 10^{-7} \text{ mol}^3 \text{ dm}^{-9} < K_{\text{sp}} \text{ of Ca(OH)}_2 \end{aligned}$$

=> Ca(OH)₂ ppt not formed

Ans: B

Magnesium hydroxide	Calcium hydroxide
Precipitated	Not precipitated

- 14.** E_{red} (Cu²⁺/Cu) = +0.34V (less positive, Cu oxidised to Cu²⁺)
Anode: negatively charged
E_{red} (Fe³⁺/Fe²⁺) = +0.77 (more positive, Fe³⁺ reduced to Fe²⁺)
Cathode: positively charged

Electrons flow from anode (Cu) to cathode (Pt)

Ans: B Electrons flow from the copper electrode to the platinum electrode.

15. $Q = It = 350\,000 \times 8 \times 60 \times 60 = 1.008 \times 10^{10} \text{ C}$
 $n = Q / F = 1.008 \times 10^{10} / 96500 = 1.04 \times 10^5 \text{ mol}$
 1 mole of Al^{3+} required 3 moles of electrons to discharge.
 No of moles of Al^{3+} discharged = $(1.04 \times 10^5) / 3 = 3.48 \times 10^4 \text{ mol}$
 Mass of Al discharged = $3.48 \times 10^4 \times 27 \times 10^{-3} = 940 \text{ kg}$

Ans: D 940 kg

16. Ionic oxides are basic, covalent oxides are acidic and ionic oxides with covalent character are amphoteric.

Option **A**: Both SiO_2 and P_4O_6 are acidic while Al_2O_3 amphoteric.

Option **B**: MgO is basic while SiO_2 and SO_3 are basic.

Option **C**: Na_2O is basic, SiO_2 is acidic and P_4O_{10} is acidic.

Option **D**: SO_2 , SiO_2 and P_4O_{10} are all acidic.

Ans: C Na_2O Al_2O_3 P_4O_{10}

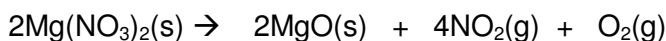
17. Option **A**: Ionic radii should decrease across the period from Groups I to III, and Groups V to VIII with the anionic radii bigger than the cationic radii.

Option **B**: Electrical conductivity should increase across the metals from Groups I to III, decrease for Group IV (semi-conductor) and then at almost nil for Groups V – VIII for non-metals.

Option **C**: Second ionization energy should increase across the period, with drops at Group IV and Group VII due to additional shielding effects provided by additional 3s electrons and interelectronic repulsion experienced by paired electrons in the same orbitals outweighing the increased nuclear charge respectively.

Ans: D Melting point

18. Magnesium nitrate decomposes according to the following equation:



Ans: D Nitrogen

19. Down group II, $E^\circ(M^{2+}/M)$ becomes more negative.

Hence Group II metals become more easily oxidized, reactions with H_2O occur more readily.

Ans: C

Be	Mg	Ca
No reaction with water.	Reacts readily with steam.	Reacts readily with cold water.

20. Option **A**: Down group VII, $E^\circ(X_2/X^-)$ becomes less positive. X_2 becomes more easily reduced and hence, oxidizing power decreases from Cl_2 to Br_2 to I_2 .

Option **B**: Down group VII, size of atom increases. Less effective overlap occurs between the H and the X atomic orbitals, resulting in weaker H – X bond down the group. Less energy is required to break the H – X bond, hence thermal stability decreases from HCl to HBr to HI .

Option **C**: Down the group, no of electrons in HX molecules increase. As the electron cloud of HX becomes larger and more easily polarized, stronger induced dipole-induced dipole interactions exist between molecules, requiring more energy to overcome. Boiling point increases from HCl to HBr to HI .

Option **D**: Down group VII, $E^\circ(X_2/X^-)$ becomes less positive. X^- becomes more easily oxidised and hence, reducing power increases from HCl to HBr to HI , that is, reducing power decreases from HI to HBr to HCl .

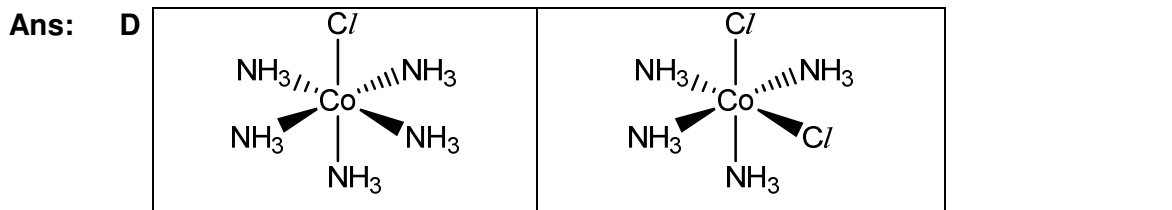
Ans: C Boiling point of HCl , HBr , HI

21. Compound **X**: 2 Cl^- for precipitation with Ag^+

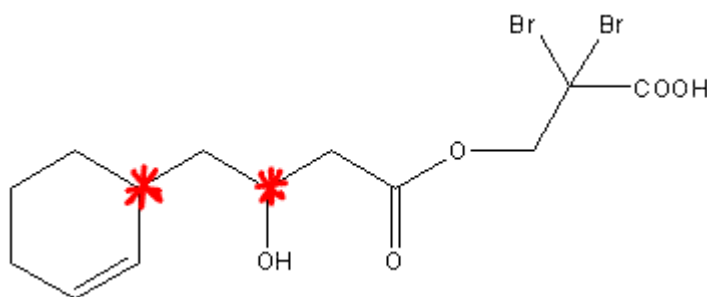
Compound **Y**: 1 Cl^- for precipitation with Ag^+ $\therefore$ Only Option **C** and Option **D** are possible answers.

Compound **Y** in Option **C** is does not have a net dipole moment.

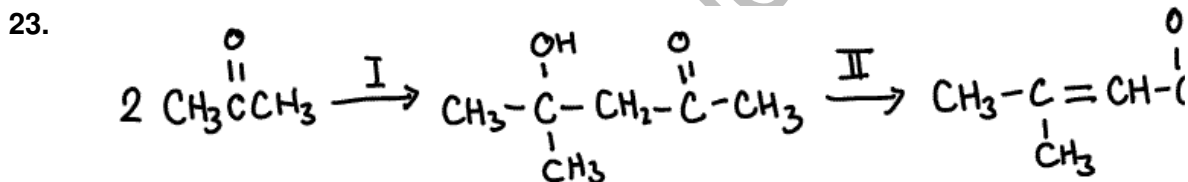
X	Y
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22. $2^2 = 4$



Ans: C 4



Ans: C $\text{CH}_3\text{COCH}=\text{C}(\text{CH}_3)_2$

24. Increasing pKa $\rightarrow$ more basic/less acidic

I: Presence of electron donating ethyl group on benzene ring

II: Presence of amine group so most basic

III: Presence of electron withdrawing fluorine on benzene ring so most acidic

Order of increasing basicity

III < IV < I < II

Ans: D III, IV, I, II

25. Val-His

His-Val

Val-Phe

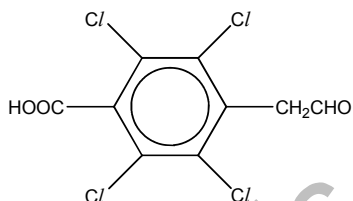
Phe-Leu

Leu-Ala

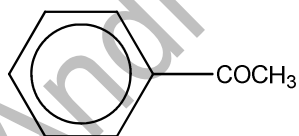
Val-His-Val-Phe-Leu-Ala

Ans: A Val-His-Val-Phe-Leu-Ala

26.	Compound	Reacts with aqueous Na_2CO_3	Brick red precipitate with Fehling's solution.	Green colouration with hot acidified $\text{K}_2\text{Cr}_2\text{O}_7$
	A	Yes	Yes	Yes
	B	Yes	No	Yes
	C	Yes	No. Fehling's solution does not react with aromatic aldehydes.	Yes
	D	No	Yes	Yes

Ans: A

27. Options **A** and **C** do not have the required structure of either R-COCH_3 or R-C(OH)HCH_3 . Option **B** is wrong because the C=O is part of an amide linkage.

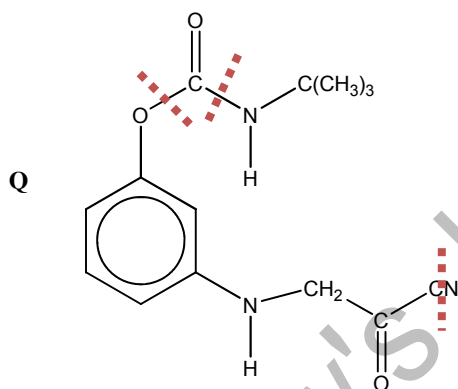
Ans: D

28.	Compound	Organic product(s)
	A	$ \begin{array}{c} \text{CH}_3 \\ \\ \text{C}=\text{O} \\ \\ \text{CH}_3 \end{array} \quad \begin{array}{c} \text{O} \quad \quad \text{O} \\ \quad \quad \\ \text{HO}-\text{C}-\text{CH}_2-\text{CH}_2-\text{C}-\text{OH} \end{array} $ Terminal double bond oxidised into CO_2 and H_2O

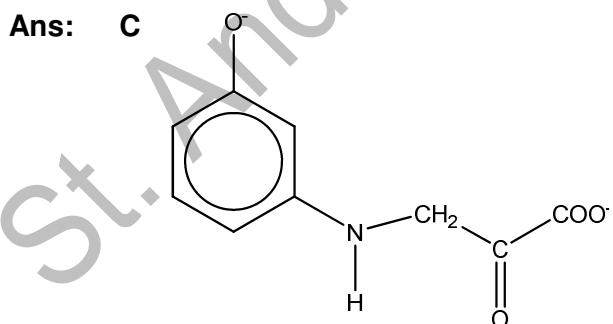
B	CH ₃ (CH ₂) ₄ COOH only. The other side forms ethandioic acid which in the presence of a strong oxidising agent is oxidised into CO ₂ and H ₂ O.
C	CH ₃ (CH ₂) ₂ COOH only. This compound is symmetrical about the double bond. Upon oxidation, both products are the same compound.
D	Benzoic acid only.

Ans: A (CH₃)₂C=CHCH₂CH₂CH=CH₂

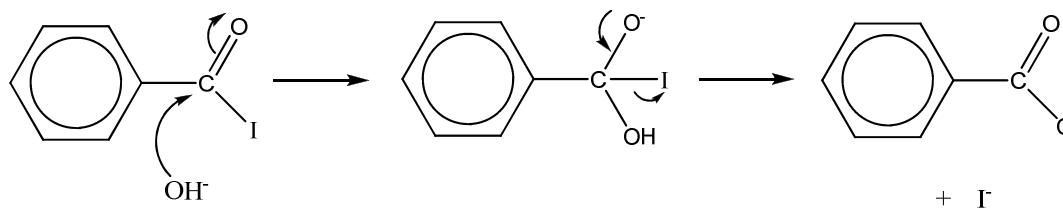
29.



Q will undergo alkaline hydrolysis as shown above. -CN becomes -COO⁻ as it is in a basic environment. The phenol formed from the hydrolysis of the ester will react with the NaOH to form the phenoxide ion.



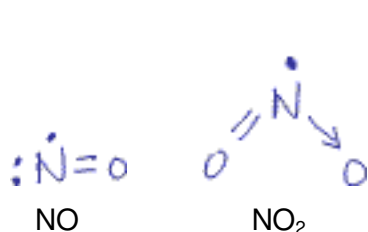
30.



The first step shows the approach of nucleophile towards a carbon which clearly has a $\delta+$ due to the electronegative atoms near it. Therefore this reaction should be classified as a nucleophilic reaction. The overall reaction shows the substitution of I by OH, making it a nucleophilic substitution overall.

Ans: D Nucleophilic substitution

31.



- 1) N atom in NO_3^- is sp^2 hybridised.
- 2) NO and NO_2 contain an unpaired electron.
- 3) NO_2 is bent and has a net dipole moment.
NO is linear and has a net dipole moment.

Ans: 1 N atom in NO_3^- is sp^2 hybridised.
 2 NO and NO_2 contain an unpaired electron.

(B)

32. $\Delta G = \Delta H - T\Delta S$

Endothermic $\Rightarrow \Delta H$ is positive

- 1) When temperature is very high and ΔS is positive, $T\Delta S > \Delta H$, ΔG is negative
- 2) When the increase in entropy is very large, $T\Delta S > \Delta H$, ΔG is negative
- 3) The change in Gibbs free energy of the reaction should be negative

Ans: 1 The temperature is very high.
 2 The increase in entropy is very large.

(B)

33. 1) Half-life for volume of X_2Y to increase from 0 to $(\frac{1}{2} \times 1)$, i.e., $0.5 \text{ dm}^3 = 20 \text{ min}$

Half-life for volume of X_2Y to increase from 0.5 to $(\frac{3}{4} \times 1)$, i.e., $0.75 \text{ dm}^3 = 20\text{min}$

Hence half-life of experiment I is constant.

- 2) In experiment I, Y is in excess, thus the rate of reaction is only dependent on X and the order of reaction w.r.t X can be found by plotting volume of X_2Y against time. Since half-life of experiment I is constant, order of reaction w.r.t X is 1.
- 3) In experiment I and II, volume of X is constant and thus the rate of reaction is only dependent on Y. Since volume of Y increases by 2 times from experiment II to I, and rate increases by 4 times, order of reaction w.r.t Y is 2.

Ans: 1 The half life of experiment I is constant.

2 The order of reaction is first order with respect to X. (B)

- 34.
- 1) No of moles of $\text{HNO}_3 = 0.00075$
 No. of moles of $\text{CH}_3\text{NH}_2 = 0.0025$
 Resultant solution consist of CH_3NH_2 and CH_3NH_3^+
 - 2) No. of moles of $\text{CH}_3\text{COOH} = 0.004$
 No of moles $\text{NaOH} = 0.005$
 Resultant solution consist of NaOH and CH_3COONa
 - 3) No. of moles of $\text{HCl} = 0.0025$
 No of moles $\text{NH}_3 = 0.0025$
 Resultant solution consist of NH_4Cl

Ans: 1 15.0 cm^3 of $0.05 \text{ mol dm}^{-3} \text{ HNO}_3$ and 25.0 cm^3 of $0.1 \text{ mol dm}^{-3} \text{ CH}_3\text{NH}_2$ (D)

35. $\text{I}_2 + 2\text{e}^- \rightleftharpoons 2\text{I}^-$ $E^\circ = +0.54 \text{ V}$

1) $\text{Cu}^{2+} + 2\text{e}^- \rightleftharpoons \text{Cu}$ $E^\circ = +0.34 \text{ V}$

$$E^\circ_{\text{overall}} = 0.34 - 0.54 = -0.20 \text{ V}$$

2) $\text{Fe}^{3+} + \text{e}^- \rightleftharpoons \text{Fe}^{2+}$ $E^\circ = +0.77 \text{ V}$

$$E^\circ_{\text{overall}} = 0.77 - 0.54 = +0.23 \text{ V}$$

3) $\text{Co}^{3+} + \text{e}^- \rightleftharpoons \text{Co}^{2+}$ $E^\circ = +1.82 \text{ V}$

$$E^\circ_{\text{overall}} = 1.82 - 0.54 = +1.28 \text{ V}$$

Ans: 2 Fe^{3+}

3 Co^{3+}

(C)

36. H-At has a much weaker bond than H-Cl and hence undergoes dissociation more readily, resulting in a larger $[\text{H}^+]$.

As iodine already disproportionate in cold dilute sodium hydroxide to yield IO_3^- and I^- as major products, astatine which is lower in Group VII would also behave likewise.

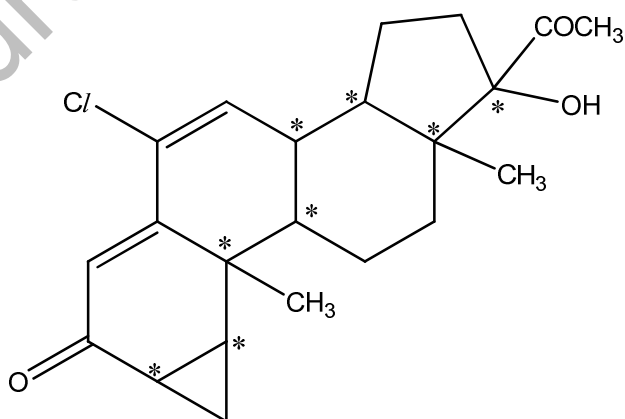
As halide proceeds down the group, it became more difficult for it to dissolve in ammonia, silver iodide was already insoluble in concentrated ammonia solution, likewise silver astatide would be insoluble in concentrated ammonia solution and hence insoluble in dilute ammonia solution.

Ans: 1 HAt is a stronger acid compared to HCl.

2 At_2 disproportionates in cold dilute sodium hydroxide to yield AtO_3^- and At^- as major products.

(B)

37. Cyproterone consists of $-\text{COCH}_3$ group which would form a yellow precipitate with alkaline aqueous iodine.



Cyproterone has got a tertiary alcohol which is resistant to oxidation.

- Ans:** 1 It forms a yellow precipitate with alkaline aqueous iodine.
 2 It possesses 8 chiral carbons. (B)

38. Naming the compound on the left (L) and the one on the right be (R)

Aqueous NaOH would neutralise L to form a homogeneous layer whereas R would form 2 separate layers.

Due to the presence of phenol in L and phenylamine in R, both compounds would form a white precipitate and decolourise aqueous bromine.

As neither L nor R possess $-\text{CH}(\text{OH})\text{CH}_3$ or $-\text{COCH}_3$ functional groups, it would not form yellow precipitate.

- Ans:** 1 Aqueous NaOH (D)

39. As the electrophile in this reaction is chlorine, there should be at least one Cl in the compound formed, with the nucleophiles (Cl^- , H_2O , and NO_3^-) completing the reaction.

- Ans:** 1 $\text{CH}_3-\underset{\text{Br}}{\text{CH}}-\underset{\text{Br}}{\text{CH}_2}$ (D)

40. Only the $-\text{COOH}$ group would react with PCl_5 and give HCl . With both the phenol and the carboxylic acid functional groups reacting with Na, it would form $\frac{1}{2}$ mole of H_2 individually. Only $-\text{COOH}$ would react with carbonate, to form $\frac{1}{2}$ mole of CO_2 .

- Ans:** 1 1 mole of A reacts with PCl_5 to give 1 mole of HCl .
 2 1 mole of A reacts with Na metal to give 1 mole of H_2 . (B)